

Systems for Socially-Challenging Information Management

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ABSTRACT

In collaborative settings, information management becomes even more challenging. In addition to the classic challenge of information management (i.e. information overload), collaboration also introduces social barriers between group members; this entails asymmetric roles, where different stakeholders have different levels of motivation, knowledge, social capital, authority, or time. To successfully manage their information, groups of people need to coordinate their needs and resources. In my thesis, I focus on building interfaces for lowering social burdens to negotiate and communicate the distribution of resources. I develop coordination systems that guide interaction for collaborative information-management tasks. I first introduce a system called Ziva, which supports interdisciplinary teams by coordinating information exchange. Then, I describe Task-Light, which aims to support information management during the help-seeking process. Lastly, I present challenges of coordinating tool preferences and a resulting system called CollaboRanger.

CCS CONCEPTS

• **Human-centered computing** → **Asynchronous editors**.

KEYWORDS

collaborative information management; online coordination; team coordination; social friction

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1 INTRODUCTION

Information management is challenging for any domain. One of the primary reasons is information overload. For example, personal information repositories such as email are overloaded and expanding daily [14]. This overloaded information (e.g., an email inbox) leads to a stressful experience for knowledge workers, makes it difficult to find information when needed, and forces users to do repetitive actions to manage their personal information space [10].

When information is shared among multiple stakeholders, an additional challenge of social barriers emerges [1]. As different stakeholders have different levels of motivation, knowledge, social

capital, authority, or time, they need to coordinate the skills and resources they have to complete the collaborative information management [4]. For example, medical doctors and data scientists form a multidisciplinary team to develop medical AI. In such teams, the team members are concerned about communication gaps stemming from mismatched expertise within the teams [11].

In my thesis, I identify social barriers in collaborative information-management task and suggest several new ways to help users to cope with it through interface support. In the following, I describe ZIVA [9], a system focusing on knowledge exchange between experts, and TASKLIGHT [8], a system leveraging help-seekers' motivation for task completion to manage information, and COLLABOR-ANGER, a system coordinating information generated by personal tools.

2 FACILITATING KNOWLEDGE SHARING IN MULTIDISCIPLINARY AI TEAMS

As successful advancement of machine learning (ML) technologies, many societal areas take advantage of ML, such as healthcare, finance, employment, and so on. However, building an ML model is still expensive and time-consuming – a common bottleneck in developing modern ML technologies is the requirement of a large quantity of labeled data. Furthermore, building a model in specialized domains poses an additional challenges; many steps in an ML development pipeline, from problem definition to feature engineering to model debugging, necessitate an understanding of *domain-specific knowledge and requirements*. Data scientists therefore often require input from domain experts to obtain labeled data, to understand model requirements, to inspire feature engineering, and to get feedback on model behavior. In practice, such knowledge transfer between domain experts and data scientists is very much ad-hoc, with few standardized practices or proven effective approaches, requires significant direct interaction between data scientists and domain experts, and is laborious process to develop guides for data scientists [11]. Building a high-quality model will inevitably require a data scientist to consult with professionals in such domains. In practice, these are often costly and frustrating iterative conversations and labeling exercises that can go on for weeks and months, which usually still do not yield output in a form readily consumable by a model development pipeline.

We proposed methods and interfaces that facilitate knowledge sharing from domain experts to data scientists for model development. We developed a domain-knowledge acquisition interface ZIVA (With Zero knowledge, How do I deVelop A machine learning model?). Instead of a data-labeling tool, ZIVA intends to provide a diverse set of elicitation methods to gather knowledge from domain experts, then present the results as a repository to data scientists to serve their domain understanding needs and to build ML models for specialized domains. ZIVA scaffolds the knowledge sharing in

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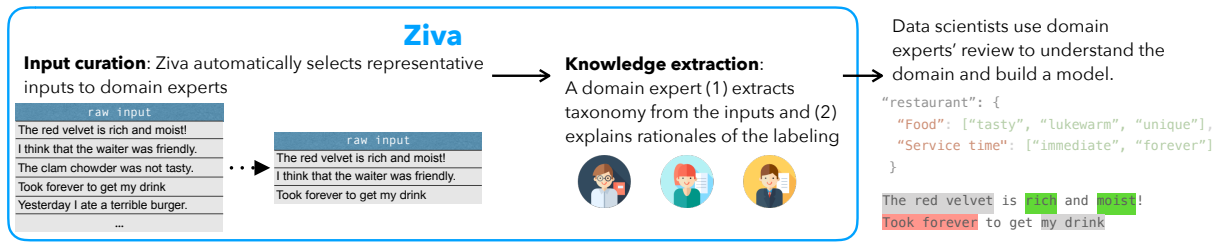


Figure 1: To facilitate domain knowledge sharing, ZIVA presents representative instances and to interfaces to review the instances to domain experts, then which will be used by data scientists.

desired formats and allows asynchronous exchange between domain experts and data scientists. It also allows flexible re-use of the knowledge repository for different modeling tasks in the domain.

Through mix-method evaluation studies, we found the completion time and subjective workload using different elicitation methods varied. Interestingly, the popular keywords based justification (bag of words) approach led to higher self-reported task success but was considered more stressful. We identified design requirements for domain knowledge-sharing tools in ML development workflow – scalability of information and lowering workload for domain experts.

3 GUIDING HELP-SEEKERS' MOTIVATION TO ALIGN WITH TASK COMPLETION

People often need to ask others to do work for them. This can happen in collaborative projects as work is divided or delegated, or can involve asking someone for a favor such as writing a recommendation letter. Any such request defines a collaborative task. Such collaborative task is often viewed through a lens of project management, in which work that must be done is assigned—by a project manager or collective decision making—to people who have to do it. However, this form of collaboration (i.e. help-seeking) entails asymmetric roles, where people who made a request do not have the power over the person who fulfills the work. The *requester* is often the one with the motivation to see it completed; however, they lack various resources, such as knowledge [9], social capital [6], authority, or time to perform the task themselves. On the other hand, the *performer* is the one capable of actually completing the work—once they have understood and committed to it. Due to such asymmetric dynamics, requesters are sometimes put in a vulnerable position and feel anxious when they seek help. Marginalized users, who frequently need to ask for help more than others, struggle even more with help-seeking.

While various collaborative task-management tools exist, requesters found that these tools were not adequate for request management despite the benefits they give for ease of tracking and efficiency; instead, they rely on email or instant messaging [6, 10]. Through our formative interviews with collaborating knowledge workers, they appreciate the clear visibility of request progress and aid for attention management, but the tools require substantial efforts to keep up-to-date status of requests from the performers' end. It is particularly problematic when the burden of this coordination

falls on the performer, even though it is the requester who is truly motivated to see the task completed.

Our approach is to identify social barriers during collaborative task management and materialize it in the life cycle. We identified that there are different types of *coordination work* done under the hood by requesters in order to see the task done:

- **Authoring well-defined and informative requests:** Requesters have difficulties on authoring initial requests to their performer since they struggle to grasp and gather information that would be useful.
- **Weathering various stages until task completion:** Collaborative tasks inherently face numerous changes and need to be adjusted accordingly, oftentimes requiring volatile coordination of non-routine intellectual work. Hence, it is natural that people have a large volume of task-related conversations over email. This leads to users juggling between discussion tools and request management tools, which can result in a loss of context and ideas. Even though collaborative task management interfaces often have commenting features, they found that comments made in the task management interface won't be as visible as simply posting on a group chat. It is because request clarification is not materialized in the interfaces or integrated in the request life-cycle.
- **Support performers' attention management:** A challenge both requesters and performers face is managing the different types of coordination work that arise when handling a request. This challenge can be multiplied if the requesters and performers are also handling numerous other requests at the same time. In some cases, however, performers' attention towards requests is not always needed. For example, if a performer asks the requester a clarification question, their attention towards the request is unnecessary until the requester responds back. On the other hand, when their attention is needed, it often requires different types of action. However, in existing task management tools, there is not much room for communicating different degree of dependency and required actions from their performers.

Motivated by these challenges, we develop a system that integrates various coordination work in task management. We provide a system that allows requesters to ask for favors and uses their motivation for completing the tasks to manage the request tracking and

The screenshot shows a web browser window with a task request form on the left and a 'Narrative Guide' message on the right.

Task Request Form:

- Task: **prepare a grant document** by **Soya**
- performers: **Cindy** (with a close icon)
- resources: **Add** (with a folder icon)
- Breadcrumbs: **Grant application > Research agenda**
- Instruction: **Decompose into smaller tasks that your performer can perform step-by-step:**
- Sub-task input: **Add**
- Status: **No subtasks for this task**

Narrative Guide:

Hi **Cindy**,

Can you help me prepare a grant document by [deadline]? I need it to be done by then because [why you need this task to be done]. This task will approximately take [How long would this take?] of your time. You can find resources of this task.

To complete the task, please do the following:
[No action item for this performer]

Thanks for the help,
Soya

Figure 2: As guidance and feedback to author informative requests, TASKLIGHT supports Narrative Guide. Narrative Guide constructs the text of a message based on the task metadata that the requester has already filled in. Using Narrative Guide, requesters can visualize how each information of requests can be used and situated for performers.

clarifications. This prevents this coordination work from falling to the performers to complete, who are already working on fulfilling the request. This way, the requesters, who are truly motivated to see the task completed, can put their effort toward a better and more efficient process for both requesters and performers.

We developed a tool, TASKLIGHT, where requesters, rather than performers, invest effort toward task coordination and guide performers through the request fulfillment. Fulfilling requests needs constant of gathering and managing information for each request. TASKLIGHT guides requesters to collect such information using Narrative Guide (Fig. 2). Next, as there are usually uncertainty and several moving parts throughout the life-cycle of requests, it is common that performers wait on and return later once requesters clarify specification. Using request with action disclosure, requesters can help managing attention and highlight of different required actions by performers in the current status of the task.

To evaluate our design, we conducted field studies. We found that TASKLIGHT allowed requesters to invest efforts toward the better process of request management. In addition, they were able to use the attention management features to better coordinate with their performers. This in turn led to better perceived success of task completion. Our results indicate that by materializing coordination work into a system, users are able to gain confidence in their help-seeking and even turn their effort on coordination work toward the better interaction.

4 COORDINATING DIFFERENCES AND PREFERENCES

Making a group decision requires understanding and processing of information or opinions from group members. If the processing is performed poorly, it is detrimental to the team's performance. However, group-decision making based on personal information is highly prone to errors, as different members produce information

in a distinct manner, and it creates challenges to merge information – starting from managing their notes, and to-dos, to managing scheduling; individuals develop their individual habits of information management. Once individuals settled into a habit, they hardly budge on it; when introduced to newer software or software updates, while individuals grasped the idea or potential benefits of adopting the new technology, they lack self-motivation to actually adopt the technology. This is because choosing particular tools to work on various tasks not only *is influenced* by personality, but also *influences* users [7]. Hence, over time, users of a tool adjust their working style in accordance with the tool that they are using. Furthermore, users prioritize finishing the immediate task quickly in the present moment despite the bigger benefits in the future. Considering all this together, individual users are unlikely to be motivated to switch tools that they handle various tasks.

Group-decision making tasks are prevalent in any groups. The tasks often span team or organizational boundaries. Traditional organizations expend significant effort in streamlining inter-group-decision making tasks by governing organizational tools [5, 13]. Recently, the advance of technology enables different forms of collaboration. Teams are more *ephemeral* and formed and dispersed around a project in a task force manner [2]. Prior work suggests that at least two challenges are heightened for ephemeral teams. First, due to the fast change of the team membership, in ephemeral teams, it is challenging to grasp and coordinate people given how people are different and team members have little understanding of each other. Second, ephemeral teams are collections of people from different organizations. Traditional teams are governed by same organizational rules and converge to their own tool suite [2, 12, 13]. Unlike traditional teams in an organization, each member in ephemeral teams is likely under different organizational rules and influences which leads to small compatibility between members [5].

In this work, we explore tool adoption in ephemeral teams. To start with, we conducted a workshop study with 11 participants

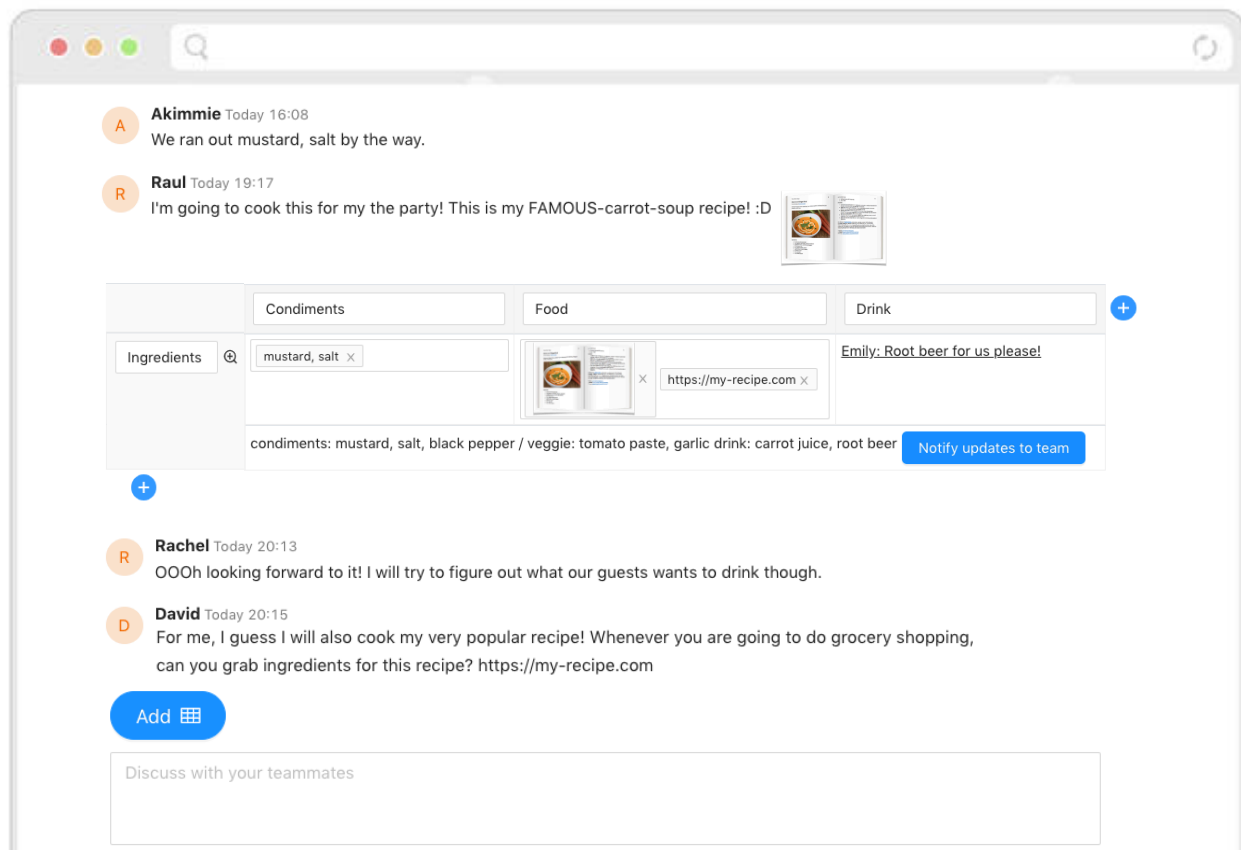
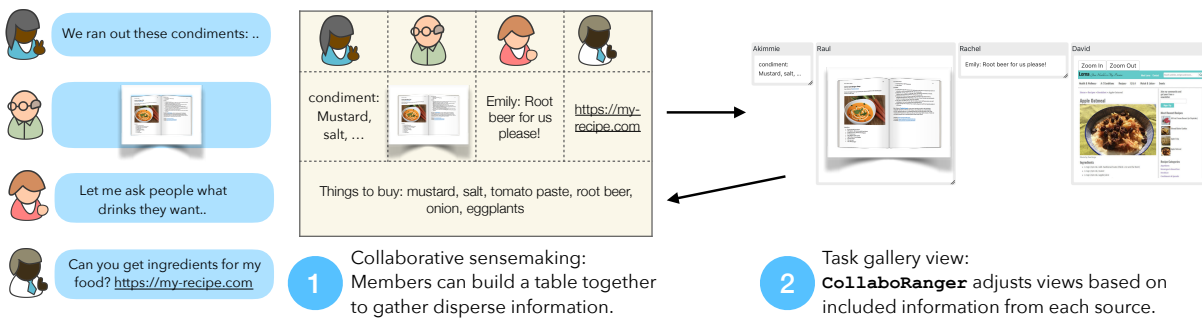


Figure 3: During the group conversation, participants can contribute to the collaborative table to make sense of information from personal tools using COLLABORANGER.

who engage in coordination with various people outside of their organizations as a part of their daily jobs (e.g., client-relation manager, admins) to understand how people cope with gaps between personal tools and group-decision making. Our findings indicate that there is careful negotiation of a way of information exchange happening in the group-decision process. However, they often “gave up” protesting for their preference due to the fear of being judged rude or inconsiderate, even if it means that the group-decision making process becomes jarring and inefficient. We also identified two classes of people in the group-decision process. The first class is a

bystander type, who engages at a minimal level. The second is a leader type, who is not necessarily the leader of the group, but takes on additional efforts to get through the group decision process.

Inspired by these challenges of coordination in ephemeral teams, we sought to design systems that encourage individuals to keep their personal tools for tasks while allow sharing and combining different sources of information from each tool more smoothly to make a group decision. Unfortunately, personal tools and groupware possess a large gap [3]. The group tool requires individuals

to totalize the way they provide responses for sensemaking; individuals with different tools for managing and formatting their information have to gather other members' responses in a uniform way. Previous work sought unified semantics of information in different tools, so information could be used across different tools. Unfortunately, computationally processing and merging information from different users is not feasible, as individuals represent the same information in vastly different ways. Although these approaches to sensemaking have been explored for individual users, little work has explored how to apply this technique to groups, which has additional challenges.

We developed COLLABORANGER, a messaging system that allows participants to gather and sensemake responses generated from individuals' preferred tools without needing to compromise or totalize individuals' tool choices of in what form the contribution should be shared. COLLABORANGER helps users to collect information from individuals' choice of tools and helps them to perform a visual comparison between them for more efficient and effective coordination; COLLABORANGER allows coordination among participants to build a table where each cell is a response linking to different tools, so that participants can see responses in various tools on one sight. Furthermore, our approach does not require a dedicated user to build the tables, allowing users of different classes (bystander vs. leader) spontaneously lead the collaborative sensemaking process.

To evaluate and inform future development of COLLABORANGER, we conducted a within-subject experiment consisting of lab and deployment studies. We recruited knowledge workers (N=19) to conduct various coordination tasks (e.g., scheduling group meetings, organizing networking, planning trips) with COLLABORANGER, and we analysed their work with COLLABORANGER as contrasted with email. Even with unfamiliar and new interfaces, COLLABORANGER users were able to finish group-coordination tasks 1.7 times faster than using email, were more informed and made more effective group decisions. They also reported that making group-decisions using COLLABORANGER demanded around 30% less mental load. Using the collaborative tables, participants were also able to send social signals, such as acknowledging other responses and contributions, which led to more social discussion and promoted engagement even for the bystanders. Participants said that they would like to adapt COLLABORANGER in their daily life, especially since the tool allows to make group decisions without forcing anyone in their team to switch their personal tools.

This work introduces a method for embracing the difference between people and letting them make informed decisions through flexible and spontaneous interfaces. More generally, it suggests a methodology to bridge gaps between personal and group tools.

5 FUTURE WORK

I'm interested in exploring system designs to expedite and facilitate information management in ad-hoc teams. As the aforementioned work discovered, this form of informal, ad-hoc nature of teams faces social barriers and conflicts to manage information together. This presents a great challenge to such teams, as team coordination and reconciliation rely on organic decisions made over time and experience [12], which ad-hoc teams do not have. Structuring and automating such team coordination could expedite it. Prior

work in social science and organization science argued that they should not be – many social interactions in groups are shown to enrich the groups and strengthen social ties and skip processes resulting in a missed opportunity [1, 12]. For example, group reflection is one of the fundamental social interactions that boost team dynamics. While such interaction is fruitful for groups, sometimes groups can not afford the time to mature or navigate the interaction. Furthermore, I found some empirical evidence that automating or structuring certain parts of team coordination can not only expedite but also improve social ties of the team; as discovered through ZIVA [9], TASKLIGHT [8] and COLLABORANGER, users were more considerate and social to their team members when they interact using the tools. Automating them might help teams bootstrap the process without losing so many benefits of them. I try to find this fine line of automating the social workflow.

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